

Abhigyan Sinha

Software Engineer

📍 Chennai, Tamil Nadu | 📞 +91 70042 55383 | ✉️ sinhaabhigyan0910@gmail.com | [🌐 LinkedIn](#) | [🐙 GitHub](#) | [🔗 LeetCode](#)

SUMMARY

Final-year B.Tech CSE (AI & ML) student at VIT Chennai with hands-on experience in deep learning, computer vision, and full-stack development. Research intern at IIT Jodhpur building CNN and BiLSTM models for real-time sign language recognition. Adept at designing and deploying production-grade AI systems and scalable web applications end-to-end.

EXPERIENCE

Research Intern – Deep Learning & Computer Vision

May 2025 – Jul 2025

IIT Jodhpur, Rajasthan (Remote) · Under Dr. Gaurav Harit, Dept. of CSE

[\[Internship Certificate\]](#)

- Built a custom CNN for static ASL classification and a hybrid 3D CNN-BiLSTM for word-level sign recognition modeling complex spatio-temporal features from video data (WLASL, IPN Hand datasets).
- Implemented a TimeDistributed CNN-BiLSTM for real-time continuous gesture stream processing and an attention-based optical flow network for enhanced dynamic gesture motion analysis.

Marketing & Sponsorship Lead

Mar 2025 – Apr 2026

Enactus VIT Chennai

- Led sponsorship acquisition and marketing campaigns as team Lead; drove cross-functional collaboration for social entrepreneurship initiatives over 2+ years.

PROJECTS

AuraFi | *Next.js, React, TypeScript, Tailwind CSS, PostgreSQL, Prisma, Plaid API, NextAuth.js*

- Full-stack personal finance platform featuring real-time bank sync via **Plaid API**, net worth tracking, budget management with visual analytics, transaction categorisation, and multi-currency support.

VisionFlow | *React, Vite, Node.js, Express, WebSockets, PostgreSQL (Supabase), JWT, HTML5 Canvas*

- Real-time crowd safety analytics platform streaming live telemetry at **60 FPS** via WebSockets; automated threshold-based anomaly detection with full incident lifecycle management and historical analytics.

StoryGen | *React, Node.js, LLM Integration*

- AI-powered creative writing app generating structured multi-chapter narratives from user prompts; includes personal library management and PDF/Word export functionality with full user authentication.

TrustFedSkin | *Python, PyTorch, EfficientNet-B0, Federated Learning (FedAvg/FedProx), CORAL Domain Adaptation, MC Dropout*

- Trustworthy AI pipeline for **4-class skin cancer classification** across 5 datasets (HAM10000, ISIC2019, BCN20000); achieved **96.9% accuracy** & **0.94 uncertainty AUROC** on external validation.
- Integrated Federated Learning, CORAL domain adaptation, and selective referral (MC Dropout + Temperature Scaling) with 98.85% accuracy at 94.4% coverage.

GreenGuard | *React 19, Node.js, Express, WebSockets, PostgreSQL, Gemini Multimodal API, OpenCorporates API*

- AI-powered greenwashing detection platform using **Google Gemini multimodal AI** for product image OCR & FTC compliance analysis, ESG report scoring, and corporate network mapping via OpenCorporates.

EDUCATION

B.Tech – Computer Science Engineering (AI & ML)

2023 – 2027

Vellore Institute of Technology, Chennai

CGPA: **8.74** / 10.0

Class XII – CBSE Board

2023

Dr. D Y Patil Pushpalata Patil International School, Patna

91.2%

TECHNICAL SKILLS

Languages

Python, TypeScript, JavaScript, Java, C++

AI / ML

Deep Learning, CNNs, BiLSTM, 3D CNN, Federated Learning, Transfer Learning, Optical Flow, Uncertainty Estimation (MC Dropout, Temperature Scaling), Computer Vision

Frameworks

PyTorch, TIMM, EfficientNet, React, Next.js, Node.js, Express.js, Prisma, Tailwind CSS

Tools & Databases

PostgreSQL, Supabase, Git, WebSockets, JWT, REST APIs, Docker, Vercel, Oracle Cloud, AWS

APIs

Google Gemini, Plaid, OpenCorporates, Open Food Facts

ACHIEVEMENTS & CERTIFICATIONS

- **Winner, VITISH 2025** – Among the 50 teams out of over 900 participating teams selected to represent VIT Chennai at the National Smart India Hackathon (SIH) 2025.
- [Neural Networks and Deep Learning](#) – Coursera (deeplearning.ai) | [Beginning C++ Programming – Beginner to Beyond](#) – Udemy
- Oracle Cloud Infrastructure Foundations – Oracle | [Generative AI for Executives](#) – AWS Skill Builder